

An EM–Bayesian GVAR: A Unified, Estimable, Real-Time Global Forecasting System

(Short paper outline — with first out-of-sample results)

M. Farkas

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Abstract

We integrate four estimation ingredients that the literature has so far deployed separately—(i) hierarchical Bayesian shrinkage (Giannone et al., 2019), (ii) EM data-completion under pervasive, ragged and pandemic-driven missingness, (iii) endogenous cross-country coupling with an *endogenous* dominant unit, and (iv) a fiscal stock–flow block—into one deployable global forecaster, the *EM–Bayesian GVAR*. The object is estimable as a single fixed point and solvable in real time: per-bloc Bayesian VARs ($p=3$) are coupled through trade-weighted “star” regressors and US-determined global commons, solved at run time as $x = Mx + d$. Making the United States endogenous yields global spillover channels a frozen-anchor GVAR cannot produce. The paper is method-led: the machinery is the engine, global forecasting is the payoff. In a recursive out-of-sample race over 56 origins (2010Q1–2024Q4) against eight benchmarks, scored by CRPS and a Model Confidence Set (MCS), the EM–Bayesian GVAR does *not* win the point race—AR(p) and a dynamic factor model lead—but it sits *inside* the MCS on the nominal block (inflation at every horizon, the short rate at $h \leq 4$), *decisively beats* the textbook OLS GVAR, and its endogenous-US variant weakly dominates the frozen one. A US +100 bp conditional projection exposes a non-trivial RoW→US feedback channel the frozen anchor structurally omits, though its magnitudes (a multi-period conditional projection, not a causal impulse response) await a validation pass. The draw-based densities are over-confident, so CRPS—not a parametric log-score—carries the density conclusions.

1 Introduction

Global VARs (Pesaran et al., 2004; Dees et al., 2007; Chudik and Pesaran, 2016) deliver cross-country structure but are estimated with fixed weights, balanced panels, and a weakly-exogenous dominant unit. Bayesian VARs (Litterman, 1986; Doan et al., 1984; Giannone et al., 2015, 2019) add shrinkage and density forecasts; the two have been combined into Bayesian GVARs, but always estimated on the same balanced panels and with the same weakly-exogenous anchor. Pandemic-era missingness (Lenza and Primiceri, 2022; Carriero et al., 2015) and ragged real-time vintages (Giannone et al., 2008) are then patched by bolt-on cleaning *before* estimation. Our contribution is not shrinkage or coupling *per se*—both exist—but to fuse data-completion, shrinkage, and coupling into *one estimable object*: an EM fixed point whose E-step completes the missing foreign data with the Durbin–Koopman smoother (Durbin and Koopman, 2002, 2012), whose M-step re-estimates the coupled Bayesian blocs, and which makes the dominant unit endogenous—all sharing a common 2025Q4 vintage across ~ 50 economies (33 GVAR blocs plus 17 euro-area standalones).

Why one estimable object beats bolt-ons. Treating completion, shrinkage, and coupling as a single fixed point makes the imputed foreign series, the prior, and the cross-country weights

mutually consistent: the stars a bloc conditions on are the smoothed paths it helps generate. Contributions:

- A **unified EM–Bayesian GVAR** estimated as one fixed point in the completed trade-weighted stars (Algorithm 1), not a pipeline of independent steps.
- **EM completion under pervasive missingness:** COVID macro-masked / financials-observed DK imputation, ragged and unbalanced panels, weights renormalized over present partners.
- An **endogenous dominant unit:** the US carries its own RoW stars and a trade-weights row, solved via an augmented-state closed form, enabling identified global spillovers.
- A **fiscal stock–flow block** closing debt dynamics by the budget identity with SFA as the exact residual, embedded in the same system.
- A **real-time, in-browser deployable** solver ($x=Mx+d$) with a pre-shipped coupling artifact, plus two designed evaluations (forecast horse-race, spillover analysis).

2 The Model

Per-bloc Bayesian VARs. For each economy i (a GVAR bloc), stack domestic variables $y_{i,t}$ and three trade-weighted foreign “star” regressors $y_{i,t}^*$:

$$x_{i,t} = \begin{bmatrix} y'_{i,t} & y_{i,t}^* \end{bmatrix}', \quad x_{i,t} = c_i + \sum_{\ell=1}^p A_{i,\ell} x_{i,t-\ell} + u_{i,t}, \quad p=3, \quad u_{i,t} \sim \mathcal{N}(0, \Sigma_i), \quad (1)$$

with a hierarchical Minnesota/GLP prior (Giannone et al., 2019) and Gibbs/Metropolis sampling of the overall-shrinkage hyperparameter (`cg1_bvarV5.m`: λ Gamma-prior mode 0.2, μ mode 1, MH-updated on the full sample).

Trade-weighted stars. The three stars are foreign-demand, foreign-inflation and foreign-financial aggregates,

$$y_{i,t}^* = \sum_{j \neq i} w_{ij,t} z_{j,t}, \quad w_{ij,t} = \frac{\bar{w}_{ij} \mathbf{1}\{j \in \mathcal{P}_{i,t}\}}{\sum_{k \neq i} \bar{w}_{ik} \mathbf{1}\{k \in \mathcal{P}_{i,t}\}}, \quad (2)$$

where $z_{j,t} \in \{\text{ROWDEM}, \text{ROWINF}, \text{ROWFIN}\}$ and weights are *renormalized over present partners* $\mathcal{P}_{i,t}$ each quarter (unbalanced panel; the macro stars may use a wider partner set than ROWFIN, which excludes rate-less or hyperinflation partners).

Solve-time coupling and global commons. Because each bloc is affine in its pinned conditioning values (fixed $\hat{A}_i, \hat{\Sigma}_i \Rightarrow$ DK filter/smoothing is affine), the assembled cross-country system is linear:

$$x = Mx + d \quad \implies \quad x = (I - M)^{-1}d, \quad (3)$$

with x the stacked bloc scenario deltas and M the (sparse) star/common transmission map (`app/src/lib/compute/gvarSolve.ts`). Four *global commons*—oil, US 10-year yield, US Baa spread, US equity—are US-determined and pinned 1:1 into every receiving bloc’s deviation path.

Endogenous vs. weakly-exogenous dominant unit. In the classical (frozen-anchor) route the US is weakly exogenous: it has no RoW block, is estimated once, and M is block-lower with the US row absent. In the endogenous route the US carries its own ROW_{US}^* stars and a trade-weights row, so the global channel closes back onto the US and (3) runs on an augmented state. Spectral radii (from the deployed coupling artifact): $\rho(M) \approx 0.59$ (6-bloc pilot), 0.723 (24-bloc frozen anchor, $\dim \approx 1380$), 0.924 (33-bloc endogenous, $\dim \approx 2060$).

3 Estimation

The system is estimated as an EM fixed point in the completed stars: the E-step completes foreign data under missingness (DK smoother), stars are rebuilt from the completed panels, and the M-step re-estimates each Bayesian bloc; iterate until the stars stop moving.

Algorithm 1: EM–Bayesian GVAR fixed-point estimation
(pipeline/export.bvar_gvar24_loop.m)

1. **Bootstrap.** Form an initial guess of each foreign star from raw partner data; mask the COVID window 2020Q2–Q4 (3 quarters), NaN-ing *macro only* (domestic macro + ROWDEM + ROWINF); keep all financial columns and ROWFIN observed.
2. **E-step (DK completion).** For each bloc, run the Durbin–Koopman smoother inside `cgl.bvarV5` to impute the macro holes *conditioned on the observed financials*, giving a completed panel $\hat{X}_i = \mathbb{E}[X_i \mid \text{observed}]$.
3. **Rebuild stars.** Recompute $y_{i,t}^*$ from the completed partner panels via (2), renormalizing weights over present partners.
4. **M-step (Bayesian estimation).** Re-estimate each bloc (1) by GLP Gibbs on [domestic | 3 stars], updating $(\hat{A}_i, \hat{\Sigma}_i)$ and the hyperparameters.
5. **Convergence.** Stop when $\max_i \|\Delta y_i^*\|_\infty < \text{TOL} = 0.03$ (Monte-Carlo floor ≈ 0.13 excluded), else go to Step 2; cap at MAXIT = 12.

Priors and COVID treatment. The GLP hierarchical prior shrinks each bloc toward independent random walks with full-sample-updated shrinkage (Giannone et al., 2019, 2015). COVID is handled as *data-augmentation missingness* (not a volatility rescaling): macro masked, financials observed, so the smoother fills the pandemic macro holes using the contemporaneous financial signal—an explicit alternative to outlier down-weighting (Lenza and Primiceri, 2022; Carriero et al., 2015).

4 System Features

Endogenous-US augmented-state closed form. With the US endogenous, the global commons enter the state as US-driven deltas, and (3) is solved on an augmented state of dimension $\dim \approx 2060$. At that size an explicit $(I - M)^{-1}$ is prohibitive in-browser, so the solver is *inverse-free*: the Neumann fixed point $x \leftarrow d + Mx$ (matvec-only, $O(\dim^2)/\text{iteration}$) converges because $\rho(M) \approx 0.924 < 1$; the frozen-anchor route instead uses $(I - M)^{-1}$ via Smith doubling. The closed form matches the iterative solve to $< 10^{-4}$ and leaves the legacy 6-bloc path byte-identical.

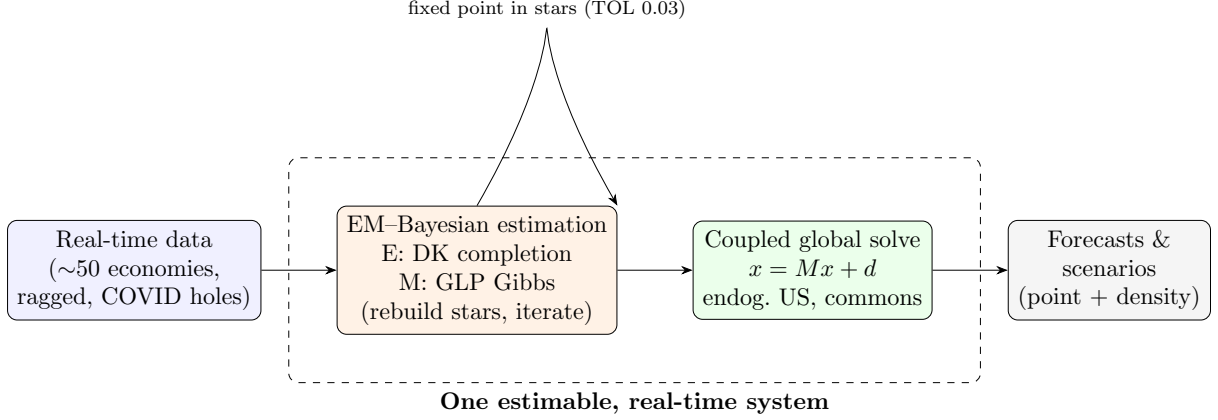


Figure 1: System schematic: data $\rightarrow$ EM-Bayesian estimation $\rightarrow$ coupled global solve $\rightarrow$ forecasts/scenarios. The dashed box is the single estimable object.

Fiscal stock-flow block. Four drivers are estimated—the revenue and primary expenditure ratios, the effective interest rate EFFI, and the stock-flow adjustment SFA—and the debt ratio is *derived* by the budget identity

$$d_t = d_{t-1} \frac{1 + i_t^{\text{eff}}/400}{1 + g_t^{\text{nom}}/100} - \frac{pb_t}{4} + \frac{sfa_t}{4}, \quad (4)$$

the standard snowball recursion (Bohn, 1998; Escolano, 2010; International Monetary Fund, 2013), with SFA the exact closing residual (`app/src/lib/compute/fiscalAccounting.ts`).

5 Application 1: An Out-of-Sample Horse-Race

Design. A recursive/expanding-window pseudo-real-time race on the 2025Q4 vintage. First origin 2010Q1, last 2024Q4; the four 2020 origins are dropped (an outlier the paper is not about), leaving 56 nominal origins. **Trade weights are rebuilt as-of each origin** (rolling 3-year window ending $\leq t_0$), the single biggest leakage control; the shipped 2021–2023 weights are forbidden in the sweep. The headline scope is the six anchor blocs (US, EA, JP, UK, CA, CN) $\times$ {growth, inflation, short rate} $\times h \in \{1, 2, 4, 8\}$. Per cell the full-sample benchmarks score ~ 55 origins; the GVAR-EM family scores ~ 51 after the short-sample ($\geq 40q$) and stability trims ($n_{\text{records}} = 33,912$). Density accuracy is scored by **CRPS** (Gneiting and Raftery, 2007) as the primary, with a 90% **Model Confidence Set** (Hansen et al., 2011) per (variable, horizon); nesting-correct tests (Giacomini and White, 2006; Clark and West, 2007; Diebold and Mariano, 1995) (with the Harvey–Leybourne–Newbold small-sample correction (Harvey et al., 1997) and Romano–Wolf step-down (Romano and Wolf, 2005) for multiplicity) underlie the stars. The benchmark set (Table 1) ladders from naive to the literature GVAR, with two deliberate ablations—`cgvar_b` (single-pass, no EM re-convergence) isolates the EM star fixed point, and `sbvar` (stars stripped) isolates the global coupling.

Result 1: AR(p) and the DFM lead the point race. The DFM wins every growth horizon (CRPS/RW = 0.75/0.84/0.86/0.84 at $h = 1/2/4/8$) and AR(p) wins inflation and the short rate at the short-medium horizons (INFL CRPS/RW = 0.63 at $h=1$; RATE CRPS/RW = 0.81 at $h=1$). We state plainly that the EM-Bayesian GVAR is *not* the point winner.

Result 2: but it is *inside* the Model Confidence Set on the nominal block. The 90% MCS (CRPS loss) places both proposed variants in the **inflation** set at *all four horizons*, and

Table 1: The nine-model OOS suite. Two proposed variants (D9), six benchmarks, plus a climatology sibling of the random walk.

Model (id)	Cross-country structure	Role / ablation
Random walk (<code>rw</code> , + <code>rw_clim</code>)	none	universal reference
AR(p) (<code>arp</code>)	none	strong univariate dynam
Single-country BVAR (<code>sbvar</code>)	none (per-bloc GLP)	value of <i>coupling</i>
Textbook OLS GVAR (<code>cgvar_a</code>)	fixed weights, exog. US, OLS	value of the Bayesian+E
Single-pass Bayesian GVAR (<code>cgvar_b</code>)	coupled, MAXIT = 1	value of the <i>EM star fix</i>
Dynamic factor model (<code>dfm</code>)	static factors	“is the GVAR structure
EM–Bayesian GVAR, frozen (<code>emgvar_frozen</code>)	coupled, exog. US	legacy anchor
EM–Bayesian GVAR, endog. (<code>emgvar_endo</code>)	coupled, endog. US	the contribution

Table 2: 90% Model Confidence Set membership (CRPS loss), six-bloc headline. ✓ = in the set (indistinguishable from the best). `emgvar_frozen` matches `emgvar_endo` except short-rate $h=1$, where it is *out*.

Block (→ <code>emgvar_endo</code>)	horizon h			
	1	2	4	8
Inflation	✓	✓	✓	✓
Short rate	✓	✓	✓	—
Growth	—	✓	✓	—

`emgvar_endo` in the **short-rate** set at $h = 1, 2, 4$ (out only at $h=8$). **Growth** is the weak block: the family is out at $h=1$ and $h=8$, in only at $h = 2, 4$ (Table 2). So on the nominal dimension the EM–Bayesian GVAR is statistically indistinguishable from the best model, despite not minimizing point loss.

Result 3: the endogenous-US design earns its keep, and the textbook GVAR is beaten decisively. In the MCS, `emgvar_endo` weakly dominates `emgvar_frozen`—the decisive case is the short rate at $h=1$, where `emgvar_endo` is *in* the set ($p = 0.150$) and `emgvar_frozen` is *out* ($p = 0.083$). The textbook OLS GVAR (`cgvar_a`) is *out* of many MCS sets and posts the worst relative CRPS in the panel (up to ≈ 2.5 at INFL $h=1$): the Bayesian shrinkage + COVID-as-missing + EM star fixed point is the demonstrable value-add over the literature GVAR—the cleanest claim the race supports.

Honest caveats. The draw-based densities are **over-confident**: the GVAR-family 90% intervals realize PIT coverage ≈ 0.73 – 0.85 (vs. the 0.90 target), worst at $h=1$, because the forecast object propagates only the within-draw shock and (β, Σ) variation from the posterior-*mean* completed state and does not integrate the DK-smoother uncertainty of the COVID-imputed states. The parametric Gaussian **log-score** is consequently *numerically broken* on these clouds (magnitudes up to $\sim 10^{16}$) and is **not reported**; **CRPS and the MCS**, being calibration-robust, carry every density conclusion.

6 Application 2: An Endogenous-US Spillover Channel

A US +100 bp *conditional projection*—hold US POLRATE at baseline +100 bp over the whole 20-quarter horizon—solved under both topologies from the *same* endogenous-staging exports (the frozen route strips the three ROW_{US}^{*} columns in-memory, so the RoW blocs and global

Table 3: US +100 bp conditional projection: peak (max-abs) response by bloc and topology, and the endogenous–frozen gap (scored space; the channel absent under the frozen anchor).

Bloc	Var	endo	frozen	gap
US	growth	+1.28	−4.37	+4.22
	infl	−1.53	−2.82	+1.30
EA	growth	+1.04	−0.95	+0.88
UK	growth	+2.14	−3.14	+2.94
JP	growth	+1.07	+1.62	−1.18
CN	infl	−0.69	+1.19	−1.29

commons are byte-identical and the gap isolates exactly one channel: US endogeneity). Under the frozen anchor the US cannot receive any RoW feedback, so its growth/inflation response is the closed-anchor own-dynamics; under the endogenous solve the global channel closes back onto the US. The **gap** = **endogenous** − **frozen** is the RoW→US feedback the frozen GVAR structurally cannot produce (Table 3): a +4.2 (growth) / +1.3 (inflation) US gap, with non-trivial, sign-varying cross-bloc gaps.

Honest caveat (not a causal IRF). This is a *multi-period conditional projection*, not a structural shock or a generalized impulse response (Pesaran and Shin, 1998): the rate path is pinned over the whole horizon. The qualitative claim—endogenizing the US opens a feedback channel the frozen anchor cannot represent—is robust; but the magnitudes are large and sign-flipping across blocs, so they **await a validation pass** (and ideally a GIRF / Diebold–Yilmaz connectedness build (Diebold and Yilmaz, 2012), the deferred Tier B) before being headlined causally. The same run doubles as a solver canary: the closed-form endogenous solve reproduces the iterative fixed point to $< 10^{-4}$ across every bloc.

7 Conclusion

The EM–Bayesian GVAR shows that Bayesian shrinkage, EM data-completion, endogenous cross-country coupling with an endogenous dominant unit, and a fiscal stock–flow block can be carried by a *single* estimable, real-time-solvable system rather than a chain of bolt-ons. The first out-of-sample evidence is honest: the system does not win the point race, but it sits inside the Model Confidence Set on the nominal block, decisively beats the textbook OLS GVAR (so the Bayesian+EM apparatus pays for itself), and its endogenous-US design weakly dominates the frozen anchor while opening a RoW→US spillover channel the latter cannot represent. The open items are scoped and stated: over-confident densities (widen the bands / integrate the smoother uncertainty) and validating the conditional-projection spillover magnitudes against a GIRF. The method is the contribution; global forecasting and the spillover channel are the payoff.

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